

Suricata IDS 탐지 시스템

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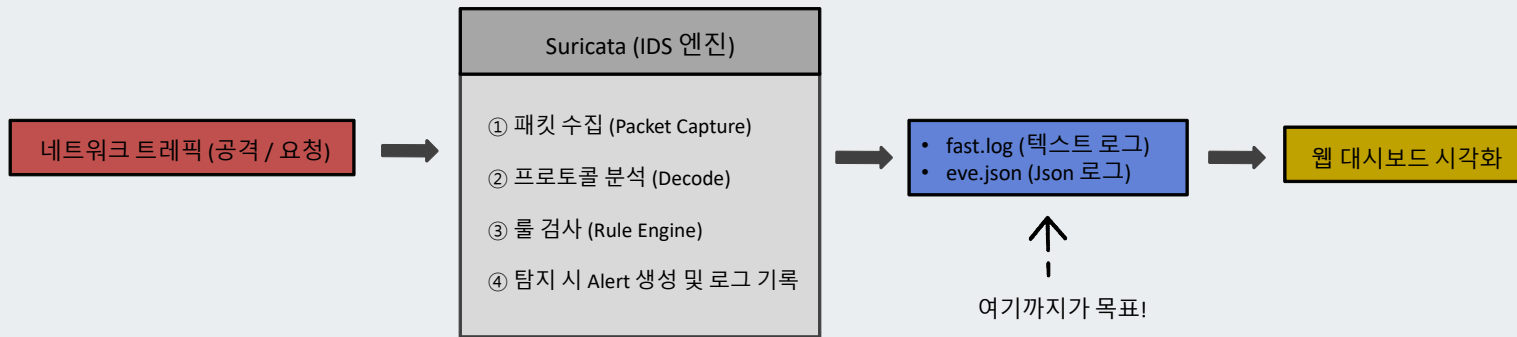
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Suricata IDS 탐지 시스템 -박혜소

- **IDS (Intrusion Detection System)** : 침입을 '탐지'하는 보안 시스템
- **Suricata** : IDS 기능을 실제로 수행하는 오픈소스 탐지 엔진



[요약]

Suricata는 IDS의 개념을 실제로 수행하는 오픈소스 엔진으로,
네트워크 트래픽을 분석하고 룰에 일치하는 행위를 탐지(Alert)하여
fast.log / eve.json 로그를 통해 결과를 기록 후 웹 대시보드에 시각화 한다

Suricata IDS 설치 - 박혜소

IDS가 설치될 환경을 최신 상태로 준비

```
ubuntu@ip:172-31-41-68: ~$ sudo apt update
To check for new updates run: sudo apt update

Last login: Tue Sep 30 06:27:37 2025 from 210.119.109.197
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip:172-31-41-68: ~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-aws-6.14.0-1013 linux-aws-6.14.0-1013 linux-headers-6.14.0-1013-aws
  linux-image-6.14.0-1013-aws linux-modules-6.14.0-1013-aws linux-tools-6.14.0-1013-aws
The following packages will be upgraded:
  bind9-dsutils bind9-host bind9-libs coreutils dkg fwupd landscope-common libc-bin libc6 libfwupd2 libipam-modules
  libipam-modules-bin libipam-runtime libipam-libs libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libselinux3-0
  libssh2-0 libxml2 linux-aws linux-headers-aws linux-images-aws linux-tools-common locales openvnm-tools
  openssl-client openssl-server openssl-sftps-server python3.12 python3.12-minimal systemd-twe-tnwb uidisks2 vim
  vim-common vim-runtime vim-tiny vim-tiny-xm
30 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
27 standard LTS security updates
```

```
sudo apt update && sudo apt upgrade -y
```

```
ubuntu@ip-172-31-41-68:~$  
User sessions running outdated binaries:  
ubuntu @ session #2: apt[1877], sshd[941,1059]  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-41-68:~$ sudo apt install -y software-properties-common jq curl python3-pip  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
software-properties-common is already the newest version (0.99.49.3).  
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).  
curl is already the newest version (8.5.0-2ubuntu0.6).  
The following additional packages will be installed:  
binutils binutils-common binutils-x86_64-linux-gnu build-essential bzip2 cpio cpp-13 cpp-13-x86_64-linux-gnu  
cpp-x86_64-linux-gnu dkms-dev fakeroot fontconfig-config fonts-dejavu-core fonts-dejavu-mono g++ g++-13  
g++-13-x86_64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86_64-linux-gnu gcc-x86_64-linux-gnu  
java-cv-jni libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-impl-perl libao3 libasan8  
libatomic1 libbinutils libc-dev-bin libc-devtools libc6-dev libc6-i386 libcrypt-dev libcrt-nobfd libcrt1  
libcrt1-dev libcdb1-perl libcpxat1-dev libfakeroot libfile-fontlock-perl libfontconfig1 libgcc-13-dev libgd3  
libgmpl1 libprofing0 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1 libhwesa0  
libim2D libintl1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc liblbs-undercore liblerc4 liblsan0  
libmpc3 libpython3-dev libpython3.12-dev libquadmath1 libsharpyuv0 libstdc++-13-dev libtiff6 libtins2  
libubsan1 libwebp7 libxpm4 linux-libc-dev lto-disabled-list make manpages-dev python3-dev python3-wheel  
python3.12-dev rcscsv-proto zlib1g-dev  
Suggested packages:  
binutils-doc profing-gui bzip2-doc cpp-doc gcc-13-locales cpp-13-doc debconf-keyring g++-multilib g++-13-multilib  
gcc-13-doc gcc-multilib autoconf automake libtool flex bison gdb gcc-doc gcc-13-multilib gdc-gcc-13-x86_64-linux-gnu
```

```
sudo apt install -y software-properties-common jq curl python3-pip
```

Suricata IDS 설치- 박혜소

오픈 소스 IDS인 Suricata를 서버에 설치

```
ubuntu@ip-172-31-41-68: ~$ sudo add-apt-repository -y ppa:oisf/suricata-stable
No VM guests are running outdated hypervisor (qemu) binaries on this host
ubuntu@ip-172-31-41-68: ~$
Repository: 'Types: deb
URLs: https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu/
Suites: noble
Components: main
Description:
Suricata IDS/IPS/NSM stable packages
https://suricata.io/
https://oisf.net/
Suricata IDS/IPS/NSM - Suricata is a high performance Intrusion Detection and Prevention System and Network Security Monitoring engine.
Open Source and owned by a community run non-profit foundation, the Open Information Security Foundation (OISF). Suricata is developed by the OISF, its supporting vendors and the community.
This Engine supports:
- Multi-Threading - provides for extremely fast and flexible operation on multicore systems.
- Multi-Tenancy - Per vlan/per interface
- Uses Rust for most protocol detection/parsing
- TLS/SSL certificate matching/logging
- JA3 TLS client fingerprinting
- JA3S TLS server fingerprinting
- IEEE 802.1ad (QinQ) and IEEE 802.1Q (VLAN) support
- VLAN support
- All JSON output/logging capability
```

`sudo add-apt-repository -y ppa:oisf/suricata-stable`

```
ubuntu@ip-172-31-41-68: ~$ sudo apt update
Hit:1 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ubuntu@ip-172-31-41-68: ~$
```

`sudo apt update`

```
ubuntu@ip-172-31-41-68: ~$ sudo apt install -y suricata suricata-update
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isa-support libevent-2.1-7t64 libevent-pthreads-2.1-7t64 libhiredis1.1.0 libhyperscan5 libluajit-5.1-common libnet1
  libnetfilter-queue1 sse3-support
The following NEW packages will be installed:
  isa-support libevent-2.1-7t64 libevent-pthreads-2.1-7t64 libhiredis1.1.0 libhyperscan5 libluajit-5.1-common libnet1
  libnetfilter-queue1 sse3-support suricata suricata-update
0 upgraded, 11 newly installed, 0 to remove and 0 not upgraded.
Need to get 7522 kB of archives.
```

`sudo apt install -y software-properties-common jq curl python3-pip`

네트워크 인터페이스 설정- 박혜소

Suricata가 감시할 네트워크 인터페이스 지정 (서버와 Suricata를 연결시키는 과정)

```
ubuntu@ip-172-31-41-68:~$ ip -o -4 route show to default | awk '{print $5}'
ens5
ubuntu@ip-172-31-41-68:~$
```

ip -o -4 route show to default | awk '{print \$5}'

← 서버의 인터페이스를 확인해줌!
-> ens5 (AWS 서버의 기본 인터페이스는 ens5임)

<Suricata 설정 편집 창>

```
GNU nano 7.2 /etc/suricata/suricata.yaml
# Suricata configuration file. In addition to the comments describing all
# options in this file, full documentation can be found at:
# https://docs.suricata.io/en/latest/configuration/suricata-yaml.html
# This configuration file was generated by Suricata 3.0.1,
# suricata-version: "3.0.1"
#
# Step 1: Inform Suricata about your network
#
# More specific is better for alert accuracy and performance
#
# address-groups:
# HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]"
# HOME_NET: "[192.168.0.0/16]"
# HOME_NET: "[10.0.0.0/8]"
# HOME_NET: "[172.16.0.0/12]"
# HOME_NET: "any"
#
# EXTERNAL_NET: "!$HOME_NET"
# EXTERNAL_NET: "any"
#
# Read 2209 lines
# Help Write Out Where Is Out Paste Execute Justify Go To Line Undo Set Mark
# Exit Read File Replace Page
```

sudo nano /etc/suricata/suricata.yaml

Suricata 인터페이스 변경

```
# Linux high speed capture support
af-packet:
- interface: eth0
# Number of packets to read ahead
# Number of packets to read ahead

# Linux high speed capture support
af-packet:
- interface: ens5
# Number of packets to read ahead
# Number of packets to read ahead
```

재시작 -> 완료

```
ubuntu@ip-172-31-41-68:~$ sudo systemctl restart suricata
ubuntu@ip-172-31-41-68:~$
```

sudo systemctl restart suricata

HOME_NET 설정 확인 - 내부 네트워크로 인식할 IP 대역 지정

```
vars:
# more specific is better for alert accuracy and performance
address-groups:
HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]"
#HOME_NET: "[192.168.0.0/16]"
```

↳ ip-172-31-41-68 (프라이빗 IP) ok

사용자/그룹 생성 및 권한 설정- 박혜소

최소 권한을 부여한 Suricata용 사용자 그룹 생성

Suricata는 네트워크 패킷을 읽고 감시하는 프로그램 -> But 패킷을 직접 감시하려면 관리자(root)권한 필요



시스템 변경 권한이 없는 Suricata 전용 최소 권한 계정 생성
(패킷 읽기 및 감시만 허용)

그룹 생성

```
ubuntu@ip-172-31-41-68:~$ sudo groupadd --system suricata
```

```
sudo groupadd --system suricata
```

사용자 생성

```
ubuntu@ip-172-31-41-68:~$ sudo useradd --system --no-create-home --gid suricata --shell /usr/sbin/nologin suricata
```

```
sudo useradd --system --no-create-home --gid suricata --shell /usr/sbin/nologin suricata
```

사용자/그룹 생성 확인

```
ubuntu@ip-172-31-41-68:~$ id suricata
uid=999(suricata) gid=988(suricata) groups=988(suricata)
```

id suricata



uid = 999(suricata) gid = 988(suricata) groups = 988

- uid = 999(suricata): 사용자 고유 ID
- gid = 988 (Suricata): 그룹 고유 ID
- groups = 988: 소속 그룹 목록 (988 그룹 하나 있음)

< 옵션 설명 >

- system: 시스템용 계정 (일반 사용자 계정 x)
- no-create-home: 홈 디렉토리 만들지 않음
- group suricata: suricata: 그룹에 속하게 함
- shell /usr/sbin/nologin: 로그인 불가능하도록 설정
- suricata : Suricata 서비스 전용 사용자 및 그룹 이름

Suricata 실행 - 박혜소

Suricata 실행(running) 확인

< Suricata 실행 결과 >

작동 중 --->

```
ubuntu@ip-172-31-41-68:~$ sudo systemctl status suricata --no-pager -l
● suricata.service - Suricata IDS/IPS/NSM/FW daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Drop-In: /etc/systemd/system/suricata.service.d
            └─override.conf
   Active: active (running) since Fri 2025-10-10 04:26:43 UTC; 2s ago
     Docs: man:suricata(8)
            man:suricatasc(8)
            https://suricata.io/documentation/
   Process: 5007 ExecStartPre=/bin/rm -f /run/suricata.pid (code=exited, status=0/SUCCESS)
   Main PID: 5010 (Suricata-Main)
     Tasks: 1 (limit: 1017)
    Memory: 106.6M (peak: 106.6M)
       CDB: 2.406e
   CGroup: /system.slice/suricata.service
            └─5010 /usr/bin/suricata --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata/suricata.pid

Oct 10 04:26:43 ip-172-31-41-68 systemd[1]: Starting suricata.service - Suricata IDS/IPS/NSM/FW daemon...
Oct 10 04:26:43 ip-172-31-41-68 systemd[1]: Started suricata.service - Suricata IDS/IPS/NSM/FW daemon.
Oct 10 04:26:43 ip-172-31-41-68 suricata[5010]: i: suricata: This is Suricata version 8.0.1 RELEASE running in SYSTEM mode
ubuntu@ip-172-31-41-68:~$
```

---> 사용자/그룹 생성 완료!

sudo systemctl status suricata --no-pager -l

룰(rule) 업데이트- 박혜소

Suricata IDS가 특정 네트워크 트래픽(SSH, HTTPS, ICMP 등) 을 탐지하도록 룰을 설정

```
ubuntu@ip-172-31-41-68:~$ sudo mkdir -p /etc/suricata/rules
ubuntu@ip-172-31-41-68:~$ sudo chown -R root:root /etc/suricata/rules
ubuntu@ip-172-31-41-68:~$ sudo chmod -R u=rwx,go=rX /etc/suricata/rules
ubuntu@ip-172-31-41-68:~$
```

룰(rule) 디렉토리 생성

- 룰(rule) 디렉토리를 생성
- 소유자를 최고 관리자(root)로 지정
- 소유자 외에는 읽기 권한만 허용

```
ubuntu@ip-172-31-41-68:~$ sudo tee /etc/suricata/rules/local.rules > /dev/null << EOF
> alert tcp any any -> any 22 (msg:"LOCAL TEST SSH SYN DETECTED"; flags:S; flow:to_server; sid:10000001; rev:1;)
> alert tcp any any -> any 443 (msg:"LOCAL TEST HTTPS SYN DETECTED"; flags:S; flow:to_server; sid:10000002; rev:1;)
> alert icmp any any -> any any (msg:"LOCAL TEST ICMP ECHO-REQUEST"; iype:8; sid:10000003; rev:1;)
> alert icmp any any -> any any (msg:"LOCAL TEST ICMP ECHO-REPLY"; iype:0; sid:10000004; rev:1;)
> EOF
ubuntu@ip-172-31-41-68:~$
```

local.rule 파일 생성 및 룰 설정

예시: alert tcp any any -> any 22 (msg:"LOCAL TEST SSH SYN DETECTED"; flags:S; flow:to_server; sid:10000001; rev:1;)

형식: <action> <protocol> <src_ip> <src_port> -> <dst_ip> <dst_port>(<옵션>)

“어떤 클라이언트(any IP, any port)가 서버(any IP)의 22번 포트(SSH)에 SYN 패킷을 보내면, Suricata가 ‘LOCAL TEST SSH SYN DETECTED’ 메시지로 경고를 발생시켜라.”

```
GNU nano 7.2 /etc/suricata/suricata.yaml
# Suricata configuration file. In addition to the comments describing all
# options in this file, full documentation can be found at:
# https://docs.suricata.io/en/latest/configuration/suricata-yaml.html
# This configuration file was generated by Suricata 0.0.1.
suricata-version: "0.0"

# Step 1: Inform Suricata about your network.

# If you are unsure of your network configuration, you can use the default
# configuration.

# more specific is better for alert accuracy and performance
address-probes:
  HOME_NET: "[192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12]"
  PHONE_NET: "[192.168.0.0/16]"
  PHONE_NET: "[10.0.0.0/8]"
  PHONE_NET: "[172.16.0.0/12]"
```

<Suricata 설정 편집 창>

```
rule-files:
- suricata.rules
##

rule-files:
- suricata.rules
- local.rules
##
## Auxiliary configuration files.
```

suricata 설정 창에 local.rule 추가

재시작 -> 완료

```
ubuntu@ip-172-31-41-68:~$ sudo systemctl restart suricata
ubuntu@ip-172-31-41-68:~$
```

sudo systemctl restart suricata

룰(rule) 업데이트 - 박혜소

Suricata IDS가 특정 네트워크 트래픽(SSH, HTTPS, ICMP 등) 을 탐지하도록 룰을 설정

```
ubuntu@ip-172-31-41-68:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
i: suricata: This is Suricata version 8.0.1 RELEASE running in SYSTEM mode
i: mpn-hs: Rule group caching - loaded: 112 newly cached: 0 total cacheable: 112
i: suricata: Configuration provided was successfully loaded. Exiting.
ubuntu@ip-172-31-41-68:~$
```

---> Suricata가 설정과 룰을 정상적으로 인식하는지 테스트 하는 과정

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

IDS Suricata 설정 완료!

공격 탐지 및 로그 확인 - 박혜소

EC2 서버에 공격 보내기

< EC2 서버 >

```
ubuntu@ip-172-31-41-68:~$ sudo tail -f /var/log/suricata/fast.log
```

sudo tail -f /var/log/suricata/fast.log
(공격이 들어오면 실시간으로 탐지로그가 뜬)

```
ubuntu@ip-172-31-41-68:~$ sudo tail -f /var/log/suricata/fast.log
18/19/2025-17:44:55.425368 [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3] [TCP] 172.31.41.68:34438 -> 91.189.91.83:80
18/19/2025-17:44:55.425368 [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification: Not Suspicious Traffic] [Priority: 3] [TCP] 172.31.41.68:34438 -> 91.189.91.83:80
18/19/2025-17:47:10.703531 [**] [1:10000002:1] LOCAL TEST HTTP SYN DETECTED [**] [Classification: (null)] [Priority: 3] [TCP] 172.31.41.68:55206 -> 169.254.169.254:80
18/19/2025-17:47:10.757876 [**] [1:10000002:1] LOCAL TEST HTTP SYN DETECTED [**] [Classification: (null)] [Priority: 3] [TCP] 172.31.41.68:55218 -> 169.254.169.254:80
18/19/2025-17:47:10.768824 [**] [1:10000002:1] LOCAL TEST HTTP SYN DETECTED [**] [Classification: (null)] [Priority: 3] [TCP] 172.31.41.68:55228 -> 169.254.169.254:80
```

fast.log

< 공격 서버 >

```
parkjiso@parkjiso-VirtualBox:~$ sudo apt update
[sudo] parkjiso 암호:
기존 :1 http://kr.archive.ubuntu.com/ubuntu noble InRelease
받기 :2 http://kr.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
```

sudo apt update

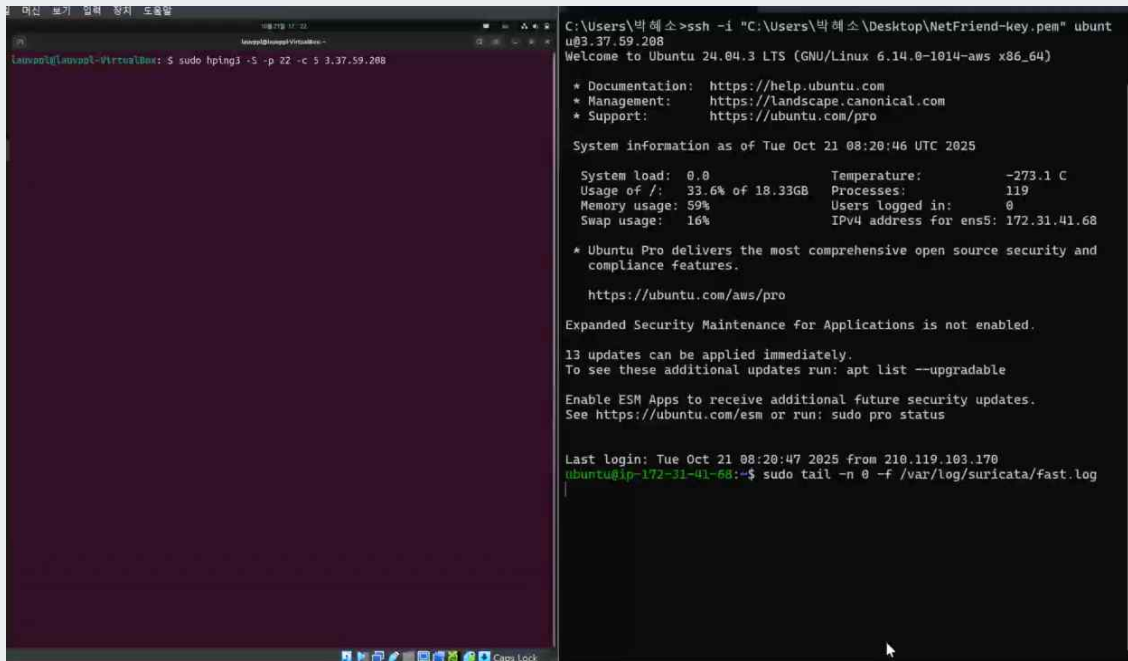
```
parkjiso@parkjiso-VirtualBox:~$ sudo apt install -y hping3
패키지 목록을 읽는 중입니다... 완료
의존성 트리를 만드는 중입니다... 완료
상태 정보를 읽는 중입니다... 완료
```

sudo apt install -y hping3

```
parkjiso@parkjiso-VirtualBox:~$ sudo hping3 -S -p 22 -c 5 3.37.59.208
[sudo] parkjiso 암호:
HPING 3.37.59.208 (enp0s3 3.37.59.208): S set, 40 headers + 0 data bytes
len=46 ip=3.37.59.208 ttl=64 id=51854 sport=22 flags=SA seq=0 win=65535 rtt=20.4 ms
len=46 ip=3.37.59.208 ttl=64 id=51855 sport=22 flags=SA seq=1 win=65535 rtt=12.9 ms
len=46 ip=3.37.59.208 ttl=64 id=51856 sport=22 flags=SA seq=2 win=65535 rtt=30.1 ms
len=46 ip=3.37.59.208 ttl=64 id=51857 sport=22 flags=SA seq=3 win=65535 rtt=19.7 ms
len=46 ip=3.37.59.208 ttl=64 id=51858 sport=22 flags=SA seq=4 win=65535 rtt=18.5 ms
```

sudo hping3 -S -p 22 -c 5 3.37.59.208

공격 탐지 및 로그 확인 - 박혜소



```
마진 보기 입력 장치 도움말
laavpo@laavpo-VirtualBox: ~
laavpo@laavpo-VirtualBox: $ sudo hping3 -S -p 22 -C 5 3.37.59.208

C:\Users\박혜소>ssh -i "C:\Users\박혜소\Desktop\NetFriend-key.pem" ubuntu@3.37.59.208
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1014-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Oct 21 08:20:46 UTC 2025

System load:  0.0           Temperature:   -273.1 C
Usage of /:   33.6% of 18.3GB Processes:    119
Memory usage: 59%          Users logged in: 0
Swap usage:  16%           IPv4 address for ens5: 172.31.41.68

 * Ubuntu Pro delivers the most comprehensive open source security and
compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Oct 21 08:20:47 2025 from 210.119.103.170
ubuntu@ip-172-31-41-68:~$ sudo tail -n 0 -f /var/log/suricata/fast.log
```

공격 탐지 및 로그 확인 - 박혜소

fast.log 로그 확인

```
ubuntu@ip-172-31-41-68:~$ sudo tail -f /var/log/suricata/fast.log
10/19/2025-17:44:55.425368  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification:
Not Suspicious Traffic] [Priority: 3] {TCP} 172.31.41.68:34438 -> 91.189.91.83:80
10/19/2025-17:44:55.425368  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification:
Not Suspicious Traffic] [Priority: 3] {TCP} 172.31.41.68:34438 -> 91.189.91.83:80
10/19/2025-17:44:55.425368  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound likely related to package management [**] [Classification:
Not Suspicious Traffic] [Priority: 3] {TCP} 172.31.41.68:34438 -> 91.189.91.83:80
10/19/2025-17:47:10.703531  [**] [1:10000002:1] LOCAL TEST HTTP SYN DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 172.31.41.68:55206 ->
169.254.169.254:80
10/19/2025-17:47:10.757876  [**] [1:10000002:1] LOCAL TEST HTTP SYN DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 172.31.41.68:55218 ->
169.254.169.254:80
10/19/2025-17:47:10.768824  [**] [1:10000002:1] LOCAL TEST HTTP SYN DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 172.31.41.68:55228 ->
169.254.169.254:80
```

```
10/19/2025-17:52:24.494327 [**] [1:10000001:1] LOCAL TEST SSH SYN DETECTED [**] [Classification: (null)] [Priority: 3] {TCP}
124.50.92.50:6265 -> 172.31.41.68:22
```

- 10/19/2025-17:52:24.494327 → 탐지 시각
- [1:10000001:1] → GID:SID:REV (여기서 SID=10000001, 우리가 지정한 룰 ID)
- LOCAL TEST SSH SYN DETECTED → 룰의 msg (탐지 메시지)
- {TCP} → 프로토콜
- 124.50.92.50:6265 → 출발지 IP:포트 (공격자 / 테스트 머신)
- -> 172.31.41.68:22 → 목적지 IP:포트 (모니터링되는 서버의 SSH)

공격 탐지 및 로그 확인 - 박혜소

eve.json 로그 확인

```
C:\Users\박혜소>ssh -i "C:\Users\박혜소\Desktop\NetFriend-key.pem" ubuntu@3.37.59.208
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1014-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct 21 08:42:30 UTC 2025

System load:  0.05      Temperature:   -273.1 C
Usage of /:   33.6% of 18.33GB   Processes:    119
Memory usage: 61%      Users logged in: 0
Swap usage:  16%        IPv4 address for ens5: 172.31.41.68

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Oct 21 08:38:24 2025 from 210.119.103.170
ubuntu@ip-172-31-41-68:~$
```

공격 탐지 및 로그 확인 - 박혜소

eve.json 로그 확인

```
{
  "timestamp": "2025-10-19T19:24:21.496814+0000",
  "flow_id": 1570851266392267,
  "in_iface": "ens5",
  "event_type": "alert",
  "src_ip": "124.50.92.50",
  "src_port": 8314,
  "dest_ip": "172.31.41.68",
  "dest_port": 22,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 10000001,
    "rev": 1,
    "signature": "LOCAL TEST SSH SYN DETECTED",
    "category": "",
    "severity": 3
  }
}
```

탐지된 이벤트(alert)

```
{
  "timestamp": "2025-10-19T19:24:21.496814+0000",
  "flow_id": 1570851266392267,
  "in_iface": "ens5",
  "event_type": "alert",
  "src_ip": "124.50.92.50",
  "src_port": 8314,
  "dest_ip": "172.31.41.68",
  "dest_port": 22,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 10000001,
    "rev": 1,
    "signature": "LOCAL TEST SSH SYN DETECTED",
    "category": "",
    "severity": 3
  }
}
```

sudo cat /var/log/suricata/eve.json | jq .

GID:SID:REV

- GID = 1 (그룹 ID)
- SID = 10000001 (우리가 지정한 룰 ID)
- REV = 1 (룰 버전)

탐지 메시지 (msg)

→ "LOCAL TEST SSH SYN DETECTED"
SSH 접속 시도(SYN 패킷) 감지

프로토콜 (proto)

→ TCP (IPv4 통신)

출발지 (공격자 / 테스트 머신)

→ 124.50.92.50:8314

목적지 (모니터링 서버 / 대상)

→ 172.31.41.68:22 (SSH 포트)

추가 정보

→ in_iface: ens5 (감시 중인 네트워크 인터페이스)

→ action: allowed (탐지만 수행, 차단 아님)

→ severity: 3 (중간 수준 경고)

공격 탐지 및 로그 확인- 박혜소

eve.json 로그 확인

탐지된 알람의 트래픽 흐름(flow)

```
{
  "timestamp": "2025-10-07T19:24:21.496814+0000",
  "flow_id": "278883320492207",
  "in_iface": "ens3",
  "event_type": "alert",
  "src_ip": "124.50.92.50",
  "src_port": 8314,
  "dest_ip": "172.31.41.68",
  "dest_port": 22,
  "proto": "tcp",
  "ip_v": 4,
  "pkt_len": 66,
  "action": "allowpass",
  "alert": {
    "engine": "allnet",
    "gid": 1,
    "signature_id": 10000001,
    "rev": 1,
    "signature": "LOCAL_TEST_SSH_KVM_DETECTION",
    "category": "",
    "severity": 9
  },
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 1,
    "pkts_toclient": 0,
    "bytes_toserver": 66,
    "bytes_toclient": 0,
    "start": "2025-10-07T19:24:21.496814+0000",
    "src_ip": "124.50.92.50",
    "dest_ip": "172.31.41.68",
    "src_port": 8314,
    "dest_port": 22
  }
}
```

```
},
"direction": "to_server",
"flow": {
  "pkts_toserver": 1,
  "pkts_toclient": 0,
  "bytes_toserver": 66,
  "bytes_toclient": 0,
  "start": "2025-10-07T19:24:21.496814+0000",
  "src_ip": "124.50.92.50",
  "dest_ip": "172.31.41.68",
  "src_port": 8314,
  "dest_port": 22
}
}
```

sudo cat /var/log/suricata/eve.json | jq .

→ 2025-10-19T19:24:21.496814+0000
탐지된 트래픽(플로우)의 시작 시각

src_ip / src_port
→ 124.50.92.50:8314
출발지 IP:포트 (공격자 또는 테스트 머신)

dest_ip / dest_port
→ 172.31.41.68:22
목적지 IP:포트 (모니터링되는 서버의 SSH 포트)

direction
→ "to_server"
트래픽이 서버 방향으로 전송되었음을 의미

pkts_toserver / pkts_toclient
→ 1 / 0
서버로 전송된 패킷은 1개, 클라이언트로 돌아온 패킷은 없음

bytes_toserver / bytes_toclient
→ 66 / 0
서버로 전송된 데이터는 66바이트. 돌아오는 바이트는 0

공격 탐지 및 로그 확인- 박혜소

eve.json 로그 확인

```
{
  "timestamp": "2025-10-19T19:24:21.496814+0000",
  "flow_id": 1570851266392267,
  "in_iface": "ens5",
  "event_type": "alert",
  "src_ip": "124.50.92.50",
  "src_port": 8314,
  "dest_ip": "172.31.41.68",
  "dest_port": 22,
  "proto": "TCP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 10000001,
    "rev": 1,
    "signature": "LOCAL TEST SSH SYN DETECTED",
    "category": "",
    "severity": 3
  },
  "direction": "to_server",
  "flow": {
    "pkts_toserver": 1,
    "pkts_toclient": 0,
    "bytes_toserver": 66,
    "bytes_toclient": 0,
    "start": "2025-10-19T19:24:21.496814+0000",
    "src_ip": "124.50.92.50",
    "dest_ip": "172.31.41.68",
    "src_port": 8314,
    "dest_port": 22
  }
}
```

시간(timestamp) → 2025-10-19T19:24:21 (UTC)

출발지(src_ip:port) → 124.50.92.50 : 8314

목적지(dest_ip:port) → 172.31.41.68 : 22

탐지명(signature) → LOCAL TEST SSH SYN DETECTED

패킷 수/크기 → 1 pkt / 66 bytes

action → allowed (탐지만 수행, 차단 X)

severity → 3 (중간 수준)

감사합니다

THANK YOU

CONTACT

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